# XRPL Validator — Daily Engineering Report · 2026-09-26

*Period: 2026-09-26 00:00 to 24:00 US Eastern (2026-09-26 04:00 to 2026-09-27 04:00 UTC); ledgers #107,240,424 to #107,262,587. Every count from the ledger and the validator's logs stops at #107,262,587. Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.*

*Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: the day's first and last ledger, from a public full-history server; signing records; Step 1 receipt records (they begin 2026-08-29); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data (carried from the original report). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); Steps 2–5 receipt records (they begin 2026-10-09); data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).*

## Status

**Normal (agreement not checked)** — coverage outside planned maintenance 100.000% (objective 99.9%), agreement with the network not checked (the reference node no longer holds these ledgers' hashes), no open engine receipt.

*Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.*

## Executive summary

- **Validator:** signed 22,155 of the 22,164 ledgers the network closed (99.959%) (agreement not checked: the ledgers' hashes are no longer held). Unsigned: 9 during planned restarts.
- **Native engine:** 0 engine receipts recorded.
- **Changes:** 1 deployment (cycle 157), 4 commits merged into main.

## Service levels

| Measure | Value |
|---|---|
| Ledgers closed by the network | 22,164 |
| Validations issued | 22,155 |
| Coverage (validations / ledgers closed) | 99.959% |
| Validations matching the network's validated ledger hash | not checked: the reference node no longer holds these ledgers' hashes |
| Unsigned during planned restarts | 9 |
| Unsigned outside maintenance | 0 |
| Coverage outside planned maintenance (objective 99.9%) | 100.000% |

| Measure | Value |
|---|---|
| Restarts | 1 |
| Basis: the hash the first peer announced as accepted (before cycle 183) | 22,155 |

Restarts:

- 16:53:48 UTC: deployment of cycle 157; validations resumed at #107,252,382 (9 ledgers unsigned between the last validation before the restart and the first after it)

*The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.*

## Native engine conformance

Ledgers with a receipt in the period (#107,240,424–#107,262,587), by step:

| Step | Ledgers with a receipt |
|---|---|
| Step 1 · native engine | 0 |
| Step 2 · records | 0 |
| Step 3 · order | 0 |
| Step 4 · peer data | 0 |
| Step 4 · validations | 0 |
| Step 5 · own hash | 0 |
| Writer | 0 |

Live counters are kept only for the current run; none exist for a past day.

## Receipt register

No receipts this day.

## Incidents

No incidents.

## Change log

### Deployments

| Cycle | Main | Deployed | Test suites | Replay windows | Post-deploy health | Soak |
|---|---|---|---|---|---|---|
| 157 | `c79ee89` | 16:53:48 UTC | n/a | n/a | 405 matched, 0 mismatched | #32 from 17:21 UTC: ended early: superseded by the next deployment (2026-09-27 12:28 UTC) |

Cycle 157 delivered: `72e281c` fix(docs): BatchV1_1 activation is now due ~2026-10-09; `c02c3b3` fix(docs): PermissionDelegationV1_1 activation is now due ~2026-10-08; `aefd9cc` fix(xrpl-node): the validator's ledger feed returns to the primary WebSocket once it is healthy; `c79ee89` fix(xrpl-node): replace the dead relay hub; back off and log failed handshakes

*Each deployment passes the unit and port test suites and replays mainnet history windows with zero mismatches before it ships, is health-checked after it starts, and then soaks for 24 hours.*

### Merged into main

| Merged | Repo | Commit | Subject |
|---|---|---|---|
| 04:23 UTC | xrpl-validator | `72e281c` | fix(docs): BatchV1_1 activation is now due ~2026-10-09 |
| 04:24 UTC | xrpl-validator | `c02c3b3` | fix(docs): PermissionDelegationV1_1 activation is now due ~2026-10-08 |
| 16:14 UTC | xrpl-validator | `c79ee89` | fix(xrpl-node): replace the dead relay hub; back off and log failed handshakes |
| 16:14 UTC | xrpl-validator | `aefd9cc` | fix(xrpl-node): the validator's ledger feed returns to the primary WebSocket once it is healthy |

## Findings fixed

No engine findings merged into main this day.

### Other engineering changes

- **fix(docs): BatchV1_1 activation is now due ~2026-10-09** ( `72e281c` , xrpl-validator) BatchV1_1 lost majority on mainnet for one voting round on 2026-09-25 (ledgers 107,227,905 to 107,228,161) and regained it at 14:46 UTC, which restarted its two-week period: activation is now due about 2026-10-09 14:46 UTC, shortly after fixBatchV1_2 (~2026-10-09 14:12 UTC), instead of ~2026-09-29. PermissionDelegationV1_1 is due about 2026-10-08 21:25 UTC. Comments and the README only.
- **fix(docs): PermissionDelegationV1_1 activation is now due ~2026-10-08** ( `c02c3b3` , xrpl-validator) Its mainnet majority restarted on 2026-09-24 (21:25 UTC), so activation is due about 2026-10-08 21:25 UTC rather than ~10-05. Comments only.
- **fix(xrpl-node): replace the dead relay hub; back off and log failed handshakes** ( `c79ee89` , xrpl-validator) xrplcluster.com has no peer port (the name leads to a web proxy): the validation relay retried it every ~25 s and never connected. The relay hubs move to relay::RELAY_HUBS as slots. Real handshakes on 2026-09-26: s1, s2 and r.ripple.com connect; hub.xrpl-commons.org answers 400 "Failed to verify session" to the same handshake; hubs.xrpkuwait.com and zaphod.alloy.ee answer 503 (full). The fourth slot alternates …
- **fix(xrpl-node): the validator's ledger feed returns to the primary WebSocket once it is healthy** ( `aefd9cc` , xrpl-validator) the validator's ledger feed moved to the next WebSocket endpoint when a connect failed, but nothing ever moved it back: reset_ws() had no caller. A fallback stint lasted until the next restart — on 2026-09-25 the reference node's xrpld upgrade left the validator's ledger feed on a public WebSocket until a warm restart.
- **feat(xrpl-node): /api/connections and /api/amendments for the ops copilot** ( `02001fb` , xrpl-validator) /api/connections reports which endpoints the validator's ledger feed and its RPC client are on ("primary", or the public fallback's URL — the primary's own URL is never echoed), how long the current fallback stint has lasted, stints and failovers since start, and each validation relay slot: the hub holding its session and for how long, sessions opened, failures in a row and the last failure. ws_sync::client() is the process's one …

- **feat(xrpl-node): vote for the pending amendments the engine is ready for** (`40d8d4a`, xrpl-validator) The vote list followed rippled 3.2.0's supported set, not the engine. It voted for XChainBridge and fixXChainRewardRounding — the XChain transactors are a blind source port with no specimen, and the rounding fix is not gated in them — and not for PermissionDelegationV1_1, which the engine implements and has verified (finding 360; devnet campaign 22, all byte-exact). James, 2026-09-26: vote accordingly. 45 votes from …

## Independence program

No independence record for UTC day 2026-09-26; records begin 2026-10-20.

## Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

## Amendment watch

*Carried over unchanged from the original report, which read it from the XRP Ledger when it was generated; the reference node no longer holds this day's ledgers.*

*Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.*

| Amendment | Majority since | Activation due |
|---|---|---|
| PermissionDelegationV1_1 | 2026-09-24 21:25 UTC | 2026-10-08 21:25 UTC |
| fixBatchV1_2 | 2026-09-25 14:12 UTC | 2026-10-09 14:12 UTC |
| BatchV1_1 | 2026-09-25 14:46 UTC | 2026-10-09 14:46 UTC |

Amendments enabled on mainnet: 94.

## Open items

None.

## Notes

*Hand-written notes on the day's XRPL work: research and operations.*

- **Soak #31: clean.** Soak #31 of build 0c76af1 (F402, the voting-ledger validation order and the vote change, deployed 2026-09-25 10:53 EDT) began at 11:20 EDT on 2026-09-25 (seed ledger 107,228,273) and ended at 11:25 EDT today with no engine receipts. The shadow compared 22,125 ledgers with no byte, result or overlay difference, and the state-hash check matched 22,278 consecutive ledgers (107,228,745 to 107,251,022) with no mismatch. The daily receipt canary, run at 12:40 EDT on 2026-09-25, was detected, so the zero count comes from a live comparison.

- **Deployment (validator connections).** Build c79ee89 replaced the live build at 12:54 EDT after its gate (1,685 suite tests and 744 flow-engine tests passing; the flow-engine run now includes campaign 25's vectors) and both mainnet replay windows (60 and 98 ledgers, zero mismatches), by warm restart: it rebuilt its state in 22.4 seconds and resumed one ledger behind the network. Its two changes concern how the validator connects, not the engine:
- The ledger subscription now returns to the reference node on its own. When the reference node cannot be reached the validator moves to a public server, and until now it stayed there until the next restart (on 2026-09-25 a warm restart moved it back). While on a public server it now checks the reference node once a minute and returns as soon as that node holds the network's latest validated ledger.
- The validation relay replaced its fourth hub, which no longer accepts peer connections. Of the hubs tested, three of Ripple's accept our handshake; two community hubs answered that they were full, and one rejected the handshake. The fourth relay now alternates between the two full hubs, and after the deployment it was accepted on its third attempt. A failed attempt is logged with its reason and retried after 10 seconds, doubling to at most 10 minutes, instead of every 25 seconds.

The health check matched 405 ledgers with no mismatches, and soak #32 began at 13:21 EDT (seed ledger 107,252,372). - **Reference node stall and RPC failover.** At 03:16 EDT the reference node stopped answering requests for about four minutes. On the same host, an unrelated maintenance command (a text search across a large directory tree) saturated the system disk, the kernel moved about 5 GB of process memory to swap on that same disk, and the node stalled until the search ended. The validator's RPC client switched to a public fallback server twice (for 10.5 and 316 seconds) and returned to the reference node on its own. Its ledger-by-ledger verification paused after ledger 107,243,458 and resumed with 107,243,459: no ledger was skipped and none mismatched. The engine was not involved. Searches of that kind no longer run on that host. - **Watchdog paging.** The ops copilot re-sent a warning every 15 minutes for as long as it stayed open, so a store-pruning warning that could not clear for days was sent 21 times in five hours. A warning is now sent once per episode, and again only when a new signal joins it; critical alerts still repeat after a cooldown (copilot commit 4233ece). The warning itself overstated the risk: it projected the fill rate from the last 24 hours, which still held the one-off writes that followed the 2026-09-25 rotation and upgrade (19.6 GB a day). Over the following day the node used about 15 GB a day (10 to 19 GB a day from hour to hour), which leaves about 40 to 50 GB free when the next rotation runs (ledger 107,436,647, about 2026-10-04), against 58 GB at the end of the previous cycle. An earlier version of this note gave 11.6 to 13.2 GB a day and 65 to 75 GB; those figures came from overnight hours only. - **fixBatchV1_2.** Its source is still not published: at 13:17 UTC there was no rippled 3.4.1 tag or release, and neither `develop` nor the 3.4 release and staging branches define the amendment. It keeps majority and activates about 2026-10-09 14:12 UTC, with BatchV1_1 following at about 14:46 UTC.

## Corrections to earlier reports

None.

## XRPL research

### Automated research — candidate amendment ideas

*Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: [iceland.halcyon-names.io](iceland.halcyon-names.io)). They are exploration leads, not proposals.*

- **Sponsored Pull-Charge Authorizations** (fees) Extend the in-flight XLS-67 Charge transaction with a ChargeAuthorization ledger object carrying sfSponsor (AccountID), sfMaxPerCharge, sfMaxTotal, an
- **Size-Proportional Reserves for DID/Credential** (fees) Add sfReserveUnits (uint32) to the DID and Credential ledger objects, computed in doApply from the serialized byte length of DIDSet's Document/URI fie
- **Domain-Scoped Iceberg Offers** (dex) Add a tfIceberg flag to OfferCreate that preflight rejects unless sfDomainID (PermissionedDEX) is also set; when present, the Offer SLE stores sfDispl

## Appendix — commits

| Committed | Merged into main | Repo | Commit | Subject |
|---|---|---|---|---|
| 09-26 04:23 | 09-26 04:23 | xrpl-validator | `72e281c` | fix(docs): BatchV1_1 activation is now due ~2026-10-09 |
| 09-26 04:24 | 09-26 04:24 | xrpl-validator | `c02c3b3` | fix(docs): PermissionDelegationV1_1 activation is now due ~2026-10-08 |
| 09-26 16:06 | 09-26 16:14 | xrpl-validator | `c79ee89` | fix(xrpl-node): replace the dead relay hub; back off and log failed handshakes |
| 09-26 16:06 | 09-26 16:14 | xrpl-validator | `aefd9cc` | fix(xrpl-node): the validator's ledger feed returns to the primary WebSocket once it is healthy |
| 09-26 19:43 | — | xrpl-validator | `02001fb` | feat(xrpl-node): /api/connections and /api/amendments for the ops copilot |
| 09-26 20:18 | — | xrpl-validator | `40d8d4a` | feat(xrpl-node): vote for the pending amendments the engine is ready for |

*Times in UTC. A commit appears on the day it was committed to main or merged into it.*